

Article

Effect of Generator Rotor Radial Deviation on the Unbalanced Magnetic Pull of 1000 MW Hydro-Generator Unit

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Abstract: The 1000 MW hydro-generator unit has huge geometrical dimensions and requires high installation accuracy. The control of its installation deviation will directly affect the installation quality and operational performance of the unit and the rotor eccentricity is one of the core problems faced in the installation and operation of the giant hydro-generator unit. This paper presents a method to calculate the asymmetric magnetic field by means of the transient finite element calculation to reveal the effect of the generator rotor radial deviation on the unbalanced magnetic pull of the unit. The effects of rotor dynamic eccentricity on the magnetic field are analyzed in the time and frequency domains, respectively. The results show that both the DC component and the key harmonic component of the unbalanced magnetic pull increase with the increase of the eccentricity value, and this variation relationship is nearly linear. When the eccentricity is no more than 0.15 mm, the unbalanced magnetic pull caused by the rotor eccentricity is calculated to be small, which is acceptable. The conclusions of this investigation can help to straighten out the requirements and standards for the installation deviation control of the large hydro-generator units.



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Keywords: 1000 MW hydro generator; installation accuracy control; eccentricity; unbalanced magnetic pull; finite element analysis

1. Introduction

In recent years, with the increase in electricity consumption and optimization of energy structure, the demand for hydropower has continued to grow. In order to meet the demand for power generation, large hydropower plants have been built one after another. Giant hydro generator units are large in size and capacity, which require very high installation accuracy and installation quality. The reasonableness of installation deviation control will directly affect the installation quality and the operating performance of the unit. The study of the mechanism of installation quality affecting unit performance will help rationalize the requirements and standards of installation deviation control.

During the installation of the generator, deviations are inevitable and small amounts of deviations are allowed. To ensure the stable operation of the generator, the range of rotor installation deviations needs to be controlled. Eccentricity and deformation of the rotor will lead to an uneven air gap, which in turn will excite an asymmetrical harmonic magnetic potential, creating an unbalanced magnetic pull (UMP). The size of the eccentricity directly affects the size and distribution of UMP. The eccentricity can be divided into static eccentricity and dynamic eccentricity according to whether the air gap distribution varies with time. For static eccentricity, the air gap distribution along the circumferential direction does not change with the rotation of the generator rotor. It is common to have the generator rotating shaft and stator in the correct position, but only the center of the rotor is shifted, this eccentricity is static eccentricity. For dynamic eccentricity, the circumferential distribution

Ignoring the B_t , The F can be reduced to the following form:

$$F_x = LR \int_0^{2\pi} \sigma(\theta, t) \cos(\theta) d\theta = \frac{LR}{2\mu_0} \int_0^{2\pi} B_n^2 \cos(\theta) d\theta \quad (15)$$

$$F_y = LR \int_0^{2\pi} \sigma(\theta, t) \sin(\theta) d\theta = \frac{LR}{2\mu_0} \int_0^{2\pi} B_n^2 \sin(\theta) d\theta \quad (16)$$

For the case of static rotor eccentricity, $\Lambda(\theta)$ is given as

$$\Lambda(\theta) = \Lambda_0 + \Lambda_s \cos\theta \quad (17)$$

Only considering the first harmonic F'_1 of f_{mm} , the expressions of F_x and F_y are shown below with the direct current (DC) component and rotating component.

$$F_x = \frac{F_1^2 RL \pi}{4\mu_0} [2\Lambda_0 \Lambda_s + \Lambda_0 \Lambda_s \cos(2\omega t - 2\beta)] \quad (18)$$

$$F_y = \frac{F_1^2 RL \pi}{4\mu_0} [\Lambda_0 \Lambda_s \sin(2\omega t - 2\beta)] \quad (19)$$

3. FE Model for UMP Computation

3.1. The Basic Model of the Hydropower Generator

The research object of this paper is a large hydro generator. It is a three-phase electrically excited synchronous generator with a stator outer diameter of more than 18 m, height approximating 5 m and rated power of 1000 MW. The stator bracket is used to fix the stator iron core. The rotor bracket is used to support the rotor pole. The upper and lower frame as well as the internal bearings play the role of fixed support. In this study, the effect of eccentricity on the magnetic pull is simulated. The basic dimensions and electrical parameters of the hydropower generator are shown in Table 1.

Table 1. Basic parameters of the hydropower generator.

Parameters (Units)	Values
Rated power (MW)	1000
Rated power factor	0.9
Number of poles	54
Number of slots	810
Winding connection	Wye
Parallel branches	9
Coil pitch	14
Frequency	50 Hz
Inner diameter of the stator	17,500 mm
Outer diameter of the stator	16,300 mm
Air gap thickness	49 mm
The armature winding end part extension	120 mm

The stator and rotor core is made of silicon steel sheet 50W250, and its magnetic nonlinearity is shown in Figure 1.